

Ashiq Ali Abdul Lathief

Address: Paderborn, Germany
Phone: +49 15223154652
E-Mail: ashiq.lathief@gmail.com
LinkedIn: [linkedin.com/in/ashiq-ali-abdul-lathief-266891191/](https://www.linkedin.com/in/ashiq-ali-abdul-lathief-266891191/)
GitHub: github.com/ashiqalathief
Education: M.Sc. Electrical Systems Eng., Universität Paderborn
Languages: English (C2), German (B1)



PROJECTS

February - August 2026

Master Thesis: Safe 3D Drone Navigation with Generative AI *Automatic Control Group (RAT), Universität Paderborn*

- Designed an end-to-end diffusion-policy pipeline (data collection, training, constrained inference, evaluation) on NVIDIA Isaac Lab / Isaac Sim
- Implemented and compared two diffusion denoiser architectures (Transformer and U-Net) with image conditioning via a Vision Transformer (ViT) observation encoder
- Used a depth camera for real-time obstacle detection, feeding obstacle geometry into the safety-constraint projection layer
- Tested the trained policy on the lab's physical drone platform, successfully validating obstacle-avoidance navigation

Technologies used: Python, PyTorch, NVIDIA Isaac Lab/Isaac Sim, SciPy (SLSQP), Zarr, NumPy, Vision Transformers (ViT), Diffusion models (DDPM), Gym-style RL environment, MuJoCo

October 2023 - 2024

Rescue Robot Object Handling System Development *GET Lab, Universität Paderborn*

- Deployed and programmed the Kinova robotic arm and the lab's custom robot (GETbot) for object handling and manipulation tasks
- Implemented AprilTag-based 6-DoF pose estimation on both robots, validating reliable object initialisation across two distinct hardware platforms
- Built and validated a Qt backend using an action server-client manipulation tasks

Technologies used: C++, Kinova Robotic Arm, AprilTag, Qt, ROS, Gazebo, Blender

EXPERIENCE

May 2023 - July 2026

Technical Product Data Support (Werkstudent)

Weidmüller Interface GmbH & Co. KG

Detmold, Germany

- Supported engineering and documentation tasks in an industrial environment
- Worked with PLM software for data maintenance of relay and its components
- Digital engineering using WMC for control cabinet construction
- Technical documentation using Adobe Illustrator and Microsoft Office

EDUCATION

April 2022 - Present

M.Sc. in Electrical Systems Engineering

Universität Paderborn

Paderborn

2017 - 2021

B.Tech (Honours) in Electrical and Electronics Engineering

Government Engineering College, Barton Hill

Kerala, India